

modeled based on analytic PN information, calibrated for example to *Effective One Body (EOB)*¹ results.

A BBH system is described by the following parameters [5]. The masses of two black holes are indicated with m_1 and m_2 , with $m_1 \geq m_2$, and the corresponding total mass is $M = m_1 + m_2$. The mass ratio is

$$q = \frac{m_2}{m_1} \leq 1$$

and the symmetric mass ratio is

$$\eta = \frac{m_1 m_2}{M^2}$$

The spin angular momentum $S_{1,2}$ of the binary components are defined in term of the dimensionless spin magnitude

$$a_{1,2} = \frac{c}{Gm_{1,2}^2} |S_{1,2}|$$

and the component aligned with the orbital angular momentum $\vec{L}$ is:

$$\chi_{1,2} = \frac{c}{Gm_{1,2}^2} |S_{1,2}| \cdot \hat{L}$$

with $\chi_i \in [-1, 1]$. Any misalignment between the spins and the orbital angular momentum induces relativistic precession effects, which cause the orbital plane to change its spatial orientation as the binary evolves. For simplicity, we consider only the case in which the spins are aligned (or anti-aligned) with the orbital angular momentum. So, we use a IMR waveform family for nonprecessing binaries. In this case, the waveform family is parameterized by only the total mass M , the symmetric mass ratio η and a single spin parameter:

$$\chi \equiv \frac{(1 + \delta)\chi_1}{2} + \frac{(1 - \delta)\chi_2}{2} \quad (3.2)$$

where

$$\delta \equiv \frac{m_1 - m_2}{M} \quad (3.3)$$

In order to track the evolution of the GW signal, we need a template which reconstructs the phase of the signal including the high-order terms of PN. We can understand this observing that the last term in the expression of the phase given by Eq. (1.48) in terms of v is:

$$\frac{3}{4} \left(\frac{GM_c}{c^3} 8\pi f \right)^{-5/3} = \frac{3}{128\eta} v^{-5/2} \quad (3.4)$$

¹The Effective One Body formalism is an analytical approach that aims to provide an accurate description of the coalescence of binary black holes with an arbitrary mass ratio.

This means that for small v the Newtonian phase Ψ is of the order of $v^{-5/2}$ and diverges for $v \rightarrow 0$. The P^1N corrections gives a contribution to Ψ of the order $v^{-3/2}$ that still diverges as $v \rightarrow 0$ and similarly all the contributions up to $P^{2.5}N$ diverge or stay finite in the small v limit; only starting for P^3N level we have correction which vanish as $v \rightarrow 0$. We have to demonstrate that in the case of signal we are studying the higher-order terms are finite and small respect the the term of $0th$ -order.

Following [45], the phase of the phenomenological waveform for non-precessing binaries is written, in the Fourier domain, as

$$\Psi = 2\pi f t_0 + \psi_0 + \frac{3}{128\eta v^5} \left(1 + \sum_{k=2}^7 v^k \psi_k \right) \quad (3.5)$$

where t_0 is the time of arrival of the signal at the detector, ψ_0 the corresponding phase, and v is defined in the same way as Eq. (1.53).

The terms ψ_k are phenomenological parameters depending on the physical parameters of the binary

$$\psi_k = \psi_k^0 + \sum_{i=1}^3 \sum_{j=0}^N x_k^{(ij)} \eta^i \chi^j \quad (3.6)$$

where Ψ_k^0 are the (P^2N) Fourier domain phasing coefficients for a test-particle, N is the minimum of $(3-i, 2)$, and $x_k^{(ij)}$ are phenomenological parameters tabulated below.

ψ_k	(ψ_k^0)	$x^{(10)}$	$x^{(11)}$	$x^{(12)}$	$x^{(20)}$	$x^{(21)}$	$x^{(30)}$
ψ_2	3715/756	-920.9	492.1	135	6742	-1053	-1.34·10 ⁴
ψ_3	-16 π + 113 χ /3	1.702·10 ⁴	-9566	-2182	-1.214·10 ⁵	2.075·10 ⁴	2.386 ·10 ⁵
ψ_4	15293365/508032·405 χ^2 /8	-1.254·10 ⁵	7.507·10 ⁴	1.338·10 ⁴	8.735·10 ⁵	-1.657·10 ⁵	-1.694·10 ⁶
ψ_6	0	-8.898·10 ⁵	6.31·10 ⁵	5.068·10 ⁴	5.981·10 ⁶	-1.415·10 ⁶	-1.128·10 ⁷
ψ_7	0	8.696·10 ⁵	-6.71·10 ⁵	-3.008·10 ⁴	-5.838·10 ⁶	1.514·10 ⁶	1.0891·10 ⁷

These waveforms are constructed in such a way that they match P^2N adiabatic inspiral waveforms in the test-mass limit, i.e. $\eta \rightarrow 0$. In this limit, $\psi_k \rightarrow \psi_k^0$.

We now estimate the phase defined in Eq.(3.5) for a chirp mass of $M_c = 10^{-3}M_\odot$, that is the maximum of the considered range mass, and for a frequency of $f = 120$ Hz. The choice of $f = 120$ Hz will be justified in the next section through the study of the signal-to-noise ratio.

Since for PBHs the dimensionless spin parameter is expected to be lower than 0.01 [1], we consider $\chi_{1,2} = 0.01$. Due to the definition of v , if the post-Newtonian terms are negligible for high mass and frequency the same will be true for any smaller mass and frequency.

With this choice, the PN correction amounts to:

$$\left(\sum_{k=2}^7 v^k \psi_k \right) \sim 0.01 \ll 1 \quad (3.7)$$

and then it is negligible with respect to the 0th-order term of the PN expansion. In standard scenarios, the spin of primordial black holes is expected to be very small, as explained in Sec. (2.2). However, if we consider that the spin of a black hole may evolve over the cosmic time due to phenomena such as accretion, and we assume, for example a value of $\chi = 0.9$, the PN correction in Eq. (3.7) becomes ~ 0.08 , which is still much smaller than the dominant contribution given by 0-order. As a consequence, we can use the quadrupole approximation to model the inspiral signal emitted by sub-solar mass PBH binaries.

This is an important preliminary result, because the quadrupole approximation allows to reduce the space parameter: the waveform will not depend on the full 15 parameter space for typical of solar mass CBC signals, but only on 4 parameters, i.e. the chirp mass, the reference time t_0 , the initial frequency f_0 , and the distance d . Using this reduced parameter space is very convenient, because it allows to limit the computational cost of the analysis.

3.2 Filtering and signal-to-noise ratio

The purpose of this section is to show that it is possible to limit the search to low frequencies, without significant losses of signal-to-noise ratio. This leads, at the same time, to a sensitivity improvement and a gain in terms of computational cost.

The output of any GW detector is a time series which contains the strain amplitude induced by the passage of a GW. This output will be a combination of a true GW signal and the noise

$$i(t) = h(t) + n(t) \quad (3.8)$$

$h(t)$ is the signal and $n(t)$ a stochastic variable, describing the detector noise. Ideally the noise of GW detectors can be described by a stationary and Gaussian stochastic process, although deviations from these assumptions often occur in practice.

The noise is typically much larger than the GW signal, so the fundamental problem is how we can extract the signal of the interest. The basic idea is the "filtering" procedure. In particular, the matched filtering technique is the best linear filter to extract a signal of known form swamped in stationary gaussian noise. [31] [10].

It relies on a very accurate knowledge of the signal shape and, if this is not the case, it requires a template bank covering the signal parameter space in such a way as not to degrade the sensitivity of the filter sensitivity too much due to a mismatch between the signal's waveforms and the template.

The performance of a filter can be quantified by means of the signal-to-noise-ratio, which is the ratio among the power of the filtered signal and of the noise.

The output signal $o(t)$ is the result of the convolution of $i(t)$ with the filter function